

PAPER The Consumable Centrepost: Coverage-versus-Protection Neutronics, Coil Integrity, and Fused Quench Detection for a Compact Spherical-Tokamak Breeder

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Abstract

The inboard centrepost of a compact spherical-tokamak (ST) breeder sits closest to the 14 MeV source and is the machine's dominant lifetime risk: at useful breeding coverage the high-temperature REBCO conductor reaches its irradiation limit in a small fraction of a full-power year (fpy). We treat the centrepost as a *consumable, scheduled-replacement* cartridge and quantify the architecture through four coupled analyses on the frozen Hyperion design point (configuration 22021, $\delta = -0.30$, $B_0 = 8$ T, centrepost peak field 16.84 T). Casting the inboard radial budget as a constrained optimisation, a three-dimensional OpenMC scan of the 72 cm inboard-column split (tungsten-carbide shield versus beryllium-multiplied breeding mantle) yields the coverage-versus-protection Pareto set: net tritium breeding ratio (TBR) falls monotonically from 1.196 to 1.097 as the shield grows from 5 to 40 cm, while centrepost life at the Fischer REBCO onset threshold (3.9×10^{22} n m⁻²) rises from 0.0048 to 0.194 fpy — the binding constraint is centrepost life, not breeding. The damage and cryogenic heat load fall with the shield at e-folding lengths of ~ 11 and ~ 9 cm respectively, and the life obeys an exact iso-elastic mission-dial law (life $\times$ source = const) that carries a $2.25\times$ margin to the 6.57 MA mid-current point. A wedged-sector finite-element model reproduces the analytic thick-wall (Lamé) bore hoop stress to -4.75% (625.3 versus 656.6 MPa). A fused strain-rate-plus-magnetization quench detector triggers at 0.089 s against 3.024 s for voltage detection — a protection lead of 2.935 s inside a 3.672 s MIITs budget, at fused signal-to-noise 322.5; an in-winding nitrogen-vacancy (NV) diamond magnetometer supplies the magnetization channel with a functional-life margin of $39\times$ over one cartridge life. A reliability/availability (RAMI) model of the demountable-cartridge swap gives a plant availability of 0.5825 for a dedicated outage, rising to 0.5904 when swaps are merged with scheduled maintenance — honestly below a continuous-duty reference. Together the analyses convert a lifetime showstopper into a scheduled maintenance line item with a quantified cadence and a quantified availability cost. No net-electricity claim is made: Hyperion is a breeder and materials platform, and these results describe how it survives its own neutron environment.

Keywords: spherical tokamak; centrepost; REBCO magnet; neutronics; quench detection; tritium breeding; availability; negative triangularity

Nomenclature

1 Introduction

1.1 The centrepost is the lifetime pinch-point

The spherical tokamak (ST) confines a large plasma current in a small volume behind a slim inboard column, which is precisely what makes it an attractive compact neutron source and breeder platform [1, 2, 3]. The same slimness places the centrepost conductor in the most intense fast-neutron field in the device. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet whose peak field is 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime chosen for its quiescent, edge-localised-mode-free edge [6, 7]. For a high-temperature REBCO centrepost, the critical-current onset threshold under fast fluence is reached in a small fraction of a full-power year, so treating the centrepost as a permanent component is not credible for a 14 MeV breeder. Neutron irradiation of REBCO coated conductor degrades the critical current well before it damages the plasma-facing structure [12, 13]; the magnet, not the wall, is the first thing to die.

Symbol	Definition
t_{sh}	tungsten-carbide inboard-shield thickness (cm); scan variable trading breeding coverage against centrepost protection
TBR	net tritium breeding ratio on the 14.06 MeV spectrum
fpv	full-power year, 3.156×10^7 s
$\Phi(\mathbf{r})$	fast-neutron fluence ($E > 0.1$ MeV) accumulated at the conductor
Φ_{lim}	REBCO fast-fluence critical-current onset threshold, 3.9×10^{22} n m $^{-2}$ (Fischer)
τ_{cp}	centrepost life to the fluence limit (fpv)
S_n	volumetric 14 MeV neutron source strength (s $^{-1}$)
σ_θ	azimuthal (hoop) stress in the centrepost conductor (MPa)
J	mean current density in the centrepost conductor (A m $^{-2}$)
MIITs	$\int I^2 dt$ adiabatic quench-protection budget (MA 2 s)
z_S, z_M, z_F	strain-rate, magnetization, and fused quench-precursor window statistics
NV	nitrogen-vacancy diamond magnetometer (in-winding sensor)
A (RAMI)	steady-state plant availability

We therefore adopt a *consumable-centrepost* architecture — a demountable, scheduled-replacement inboard cartridge, in the same spirit as the demountable-magnet ARC concept [3] — and answer four questions on the frozen design point. *(i)* How does inboard shielding trade against breeding coverage? *(ii)* Does the coil survive its own Lorentz loads? *(iii)* Can a quench be caught fast enough to protect a no-insulation winding? *(iv)* What maintenance cadence and plant availability does the swap imply? The contribution of this paper is to answer all four quantitatively, on one frozen configuration, with the neutronics, structural, protection, and availability analyses coupled through a single inboard radial budget, and to show that the answers convert a lifetime showstopper into a scheduled maintenance line item.

1.2 Prior art and the design context

Neutron shielding of an ST centre column is an established but unresolved design tension. Windsor and co-workers showed that tungsten borides outperform monolithic tungsten as centre-column shields by roughly an order of magnitude in flux and heating attenuation [19, 18], and the general shielding-versus-build tension for compact STs is well documented; the residual problem is that shielding the magnet steals radial space from the breeding blanket. Tritium breeding at net TBR above unity is itself a demanding requirement for a compact machine [23, 5], and the coupling between shield thickness and breeding coverage is exactly the trade this paper quantifies. On the magnet side, no-insulation (NI) REBCO technology underlies recent record fields [8, 9] and compact high-field tokamak concepts [4, 3, 11], but its slow normal-zone propagation makes quench detection a genuine protection liability [24], motivating the non-voltage, fused-sensor detector developed here [25, 26]. Damage accumulation and transmutation in the surrounding structure are quantified with the standard NRT displacement metric [20] and integrated materials-response models [21, 22], and the neutron transport itself rests on modern Monte-Carlo methods with evaluated data [14, 15] and global variance reduction [17].

1.3 Relation to the Kronos series and scope

This paper is a companion to the Hyperion breeder platform [31] and shares its configuration and neutronics basis. It sits at the hub of a cluster of companion results: the deeper fast-fluence/dpa life-limiter analysis [33], the source-strength-versus-life operating-window synthesis [34], the first-principles radiation tolerance of centrepost REBCO [35], the REBCO high-field conductor basis [36], the net-blanket-TBR fleet case [40], the dedicated fused NV-diamond quench-detection study [37], the demountable-cartridge swap-logistics and fleet-RAMI models [38], and the distinct 26.49 T burner plug coil [39] — a separate, higher-field magnet that must never be conflated with the 16.84 T breeder centrepost. Throughout, *no net-electricity claim is made*: Hyperion is a breeder and strategic-materials platform, and the results below describe only how it survives its own neutron environment. Every headline quantity is a frozen anchor in the Kronos Physics De-Risking Register [32], and the register gate identifiers (e.g. BR-L1-A13b) are given inline so each number is auditable.

2 Governing equations and the formal problem

2.1 The coverage-versus-protection problem

The inboard column has a fixed radial budget. Axis outward, it is built as a homogenised Cu/SS316 conductor ($r < 20$ cm), a tungsten-carbide shield of thickness t_{sh} , and the beryllium-multiplied breeding mantle. With a 72 cm column and a 20 cm conductor, the mantle

build is $47 - t_{\text{sh}}$ (in cm), so the shield and the mantle compete one-for-one: every centimetre of protection is a centimetre of lost breeding coverage. The design problem is therefore a scalar constrained optimisation over the single variable t_{sh} ,

$$\max_{t_{\text{sh}}} \tau_{\text{cp}}(t_{\text{sh}}) \quad \text{s.t.} \quad \text{TBR}(t_{\text{sh}}) \geq \text{TBR}_{\text{min}}, \quad t_{\text{sh}} \in [t_{\text{min}}, 47 \text{ cm}], \quad (1)$$

where τ_{cp} is the centrepost life and TBR_{min} is the breeding floor set by the fuel-cycle requirement. Both objective and constraint are monotone but opposed in t_{sh} (life rises, TBR falls), so the solution lives on the Pareto boundary; the engineering choice is where on that boundary to sit. The remainder of this section derives $\text{TBR}(t_{\text{sh}})$ and $\tau_{\text{cp}}(t_{\text{sh}})$ from transport, the coil stress from elasticity, the protection lead from the quench heat balance, and the availability from a renewal model.

2.2 Neutron transport, fluence, and life

The angular flux $\psi(\mathbf{r}, \mathbf{\Omega}, E)$ obeys the steady linear Boltzmann transport equation,

$$\mathbf{\Omega} \cdot \nabla \psi + \Sigma_t(\mathbf{r}, E) \psi = \int_{4\pi} \int_0^\infty \Sigma_s(\mathbf{r}, E' \rightarrow E, \mathbf{\Omega}' \cdot \mathbf{\Omega}) \psi(\mathbf{r}, \mathbf{\Omega}', E') dE' d\mathbf{\Omega}' + \frac{S_n(\mathbf{r})}{4\pi} \chi(E), \quad (2)$$

with macroscopic total and scattering cross-sections Σ_t, Σ_s and a monoenergetic 14.06 MeV D–T source of strength S_n and spectrum $\chi(E) = \delta(E - 14.06 \text{ MeV})$. Two functionals of the solution close the problem. The net tritium breeding ratio integrates the ${}^6\text{Li}(n, \alpha)t$ and ${}^7\text{Li}(n, n'\alpha)t$ reaction rates over the mantle and normalises to the source,

$$\text{TBR} = \frac{1}{S_n} \int_{V_{\text{mantle}}} \int_0^\infty [\Sigma_6(\mathbf{r}, E) + \Sigma_7(\mathbf{r}, E)] \phi(\mathbf{r}, E) dE dV, \quad (3)$$

with $\phi = \int_{4\pi} \psi d\mathbf{\Omega}$ the scalar flux. The fast-fluence rate at the conductor is the $E > 0.1 \text{ MeV}$ flux, and the centrepost life is the Fischer onset threshold divided by that rate,

$$\dot{\Phi}_{\text{fast}}(\mathbf{r}) = \int_{0.1 \text{ MeV}}^\infty \phi(\mathbf{r}, E) dE, \quad \tau_{\text{cp}} = \frac{\Phi_{\text{lim}}}{\max_{\mathbf{r} \in V_{\text{cp}}} \dot{\Phi}_{\text{fast}}(\mathbf{r})}, \quad \Phi_{\text{lim}} = 3.9 \times 10^{22} \text{ n m}^{-2}. \quad (4)$$

Because equations (3)–(4) are ratios of flux to source, both TBR and the flux-per-source are pure functions of geometry: they do not depend on the absolute source strength. This is the origin of the *iso-elastic mission-dial law*. Writing the operating source strength as S_n and holding the geometry fixed, the fluence rate scales linearly with S_n and the life scales inversely,

$$\dot{\Phi}_{\text{fast}} \propto S_n \implies \tau_{\text{cp}}(S_n) S_n = \text{const.} \quad (5)$$

Hence the same geometry that yields the design-point life at $I_p = 9.66 \text{ MA}$ yields exactly $\tau_{\text{cp}} \times (S_n^{\text{des}}/S_n^{\text{mid}})$ at the reduced 6.57 MA mid-current point; the frozen ratio is 2.25. The displacement-damage rate follows the same geometry-only scaling through the NRT kernel [20],

$$\dot{d}_{\text{dpa}}(\mathbf{r}) = \frac{1}{N} \int_0^\infty \phi(\mathbf{r}, E) \left[\int_0^\infty \sigma(E, T) \nu_{\text{NRT}}(T) dT \right] dE, \quad \nu_{\text{NRT}} = \frac{0.8 T_{\text{dam}}}{2E_d}, \quad (6)$$

with N the atom density, $\sigma(E, T)$ the recoil cross-section, T_{dam} the damage energy and E_d the displacement threshold; the resulting $\dot{d}_{\text{dpa}}$ and the volumetric nuclear heating both attenuate quasi-exponentially through the shield.

2.3 REBCO critical-current degradation and the fluence limit

The threshold in equation (4) is a property of the conductor, not of the geometry. Fast-neutron irradiation introduces point defects and cascade damage in the REBCO layer, and the in-field critical current I_c falls with accumulated fluence. Over the operationally relevant range the degradation is well described by an initial plateau followed by a near-exponential decline,

$$\frac{I_c(\Phi)}{I_c(0)} = \exp[-(\Phi/\Phi_0)^m], \quad \Phi_{\text{lim}} \equiv \Phi \text{ at } I_c/I_c(0) = 1 - \eta_c, \quad (7)$$

where Φ_0 and the stretch exponent m are fitted to reactor-irradiation data and η_c is the tolerable critical-current loss. Taking the Fischer onset value $\Phi_{\text{lim}} = 3.9 \times 10^{22} \text{ n m}^{-2}$ [12] as the retirement criterion is conservative: it is the fluence at which the in-field I_c first departs measurably from its unirradiated value, well before the steep part of equation (7). The first-principles basis of Φ_{lim} — chain-oxygen-vacancy formation and displacement cascades in YBCO — is developed in the companion radiation-tolerance study [35]; here it enters only as the scalar retirement fluence.

2.4 Coil structural integrity: the Lamé hoop-stress bound

The centrepost carries the toroidal-field current at the smallest radius in the machine, so its structural limit is set by its own self-field body force rather than by any external load. For an axisymmetric conductor of outer radius a carrying total current I at mean current density $J = I/\pi a^2$, Ampère's law gives the enclosed azimuthal field $B_\theta(r) = \mu_0 J r/2$, and the Lorentz body force density is radial and inward-compressing,

$$\mathbf{f} = \mathbf{J} \times \mathbf{B} = -\hat{\mathbf{r}} \frac{\mu_0 J^2 r}{2}. \quad (8)$$

Treating the conductor as a thick-walled cylinder in equilibrium, the radial and azimuthal (hoop) stresses obey the elasticity balance $d\sigma_r/dr + (\sigma_r - \sigma_\theta)/r + f_r = 0$ closed by the plane-strain compatibility relation [30]. Integrating with a traction-free outer surface and a solid core yields the classical Lamé-type solution whose peak is the membrane hoop stress at the bore,

$$\sigma_\theta^{\max} = \frac{\mu_0 J^2 a^2}{4} g(\nu), \quad g(\nu) = \frac{3+\nu}{4} + (\text{winding-pack correction}), \quad (9)$$

with ν the effective Poisson ratio of the winding pack and $g(\nu)$ an $O(1)$ Poisson/geometry factor. The monotone radial profile — peak at the bore, falling outward — identifies the bore membrane hoop stress as the single binding structural quantity, and equation (9) is the closed-form target that the finite- element model must reproduce (§4.4).

2.5 Quench heat balance, the MIITs budget, and fused detection

A no-insulation REBCO centrepost has a large temperature margin but a very slow normal-zone propagation velocity, so a developing hot spot can reach a damaging temperature before it generates a measurable resistive voltage [24]. In the adiabatic hot-spot limit the local energy balance equates Joule heating to the enthalpy rise of the conductor,

$$J(t)^2 \rho(T) = \gamma C_p(T) \frac{dT}{dt}, \quad (10)$$

with electrical resistivity ρ , mass density γ and specific heat C_p . Separating variables and integrating from the operating temperature T_{op} to a material limit T_{max} defines the protection budget in the MIITs (mega-amp-squared-seconds) form,

$$\int_0^{t_q} I(t)^2 dt = A_{\text{cond}}^2 \int_{T_{\text{op}}}^{T_{\text{max}}} \frac{\gamma C_p(T)}{\rho(T)} dT \equiv \text{MIITs}, \quad (11)$$

where A_{cond} is the conductor cross-section; the frozen budget is $3.672 \text{ MA}^2 \text{ s}$. A detector must fire while the accumulated $\int I^2 dt$ is safely below this bound. We fuse two non-voltage precursor channels: a fibre-Bragg *strain-rate* statistic z_S reading the mechanical signature of local current redistribution [25], and a *magnetization* statistic z_M reading the collapse of the screening current [26]. Each is a matched-template z -score over the precursor window. Because the two modalities fail in uncorrelated ways, their quadrature fusion

$$z_F = \frac{z_S + z_M}{\sqrt{2}}, \quad \text{SNR}_F^2 = \text{SNR}_S^2 + \text{SNR}_M^2, \quad (12)$$

carries a window statistic whose signal-to-noise ratio adds the single-channel ratios in quadrature. The detection lead is the difference between the fused trigger time and the voltage-detection time, and must be positive with margin inside the MIITs budget of equation (11).

2.6 Availability of a scheduled-swap cartridge

Because the centrepost has a finite life τ_{cp} , the plant alternates between an uptime interval $T_{\text{up}} = \tau_{\text{cp}}$ and a swap downtime T_{sw} . In the renewal (alternating-process) model the steady-state availability is

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} = \frac{T_{\text{up}}}{T_{\text{up}} + T_{\text{sw}}} \cdot \Pi_{\text{other}}, \quad (13)$$

where $\Pi_{\text{other}} < 1$ folds in all other scheduled and unscheduled outages. Merging the centrepost swap with an otherwise-scheduled maintenance window removes the swap's dedicated contribution to the denominator: if a fraction β of the swap downtime overlaps existing outages, the effective downtime is $T_{\text{sw}}(1 - \beta)$, which is the single largest controllable lever on A . Equations (1)– (13) are the formal problem; the rest of the paper evaluates them on the frozen configuration.

3 Computational methodology: the NVIDIA GPU HPC campaign

Every quantity in this paper was produced on the Kronos NVIDIA GPU HPC campaign — Kronos-owned A100/H100-class clusters for offline multi-physics Monte-Carlo and structural sweeps, with bare-metal nodes for deterministic, jitter-free integration of the protection and availability models. The campaign runs the full neutron-transport and structural sweeps that recompute the breeding and centrepost-integrity envelopes; the same frozen result set is re-derivable from the open register deposit. Four solvers were run for this paper.

3.1 Neutron transport (*OpenMC*)

The coverage-versus-protection trade (gate BR-L1-A13b) was computed with OpenMC 0.15.3 [14] on the FENDL-3.2 fusion library, with cross-check runs on ENDF/B-VIII.0 [15]. The geometry is the validated coupled three-dimensional build — elliptical plasma/SOL/first-wall/blanket/vessel tori at $R_0 = 120$ cm, $a = 48$ cm, $\kappa = 2.0$ — with the inboard column resolved, axis outward, as homogenised Cu/SS316 conductor ($r < 20$ cm), the tungsten-carbide shield of thickness t_{sh} , and the beryllium-multiplied breeding mantle ($f_{\text{Be}} = 0.80$, ^6Li enrichment 0.90). A uniform 14.06 MeV D–T volume source drives the transport of equation (2). The estimators of equations (3)–(6) are tallied on a cylindrical mesh over the inboard column. *Variance reduction.* Because the conductor sits deep behind the shield, an analogue random walk starves the tally; we generate MAGIC weight windows [17] from a first coarse pass and apply them on a cylindrical mesh, cutting the conductor fast-flux relative error to 0.01–0.03 at 4×10^5 source particles. *Convergence and verification.* The full-machine tally (gate BR-L1-A5, 1.2×10^8 histories) closes the neutron balance to leakage 0.121 and reproduces the calibration anchor bit-consistently: net TBR 1.0537, conductor fast flux 6.21×10^{11} n cm $^{-2}$ s $^{-1}$, and life 0.199 fpy at the reference split. The mid-to-design life ratio 2.25 matches the iso-elastic law of equation (5) exactly, an independent check on the source normalisation. A CAD-faithful STEP-to-DAGMC pathway reproduces the same net TBR in the companion neutronics study [42].

3.2 Coil finite-element model

The bore hoop stress of equation (9) was solved (gate 02) with a wedged-sector finite-element model of the inner turn: a periodic angular sector of the winding pack under the self-field body force of equation (8), plane-strain in the axial direction, with an effective winding-pack Poisson ratio and traction-free outer boundary. The mesh is refined toward the bore, where the membrane hoop stress peaks; convergence is declared when the bore stress changes by $< 0.5\%$ under a further halving of the element size. The solution is verified against the closed-form Lamé bound as an analytic manufactured reference (§4.4).

3.3 Quench transient and detector statistics

The adiabatic hot-spot integral of equations (10)–(11) was integrated (gate 03) with temperature-dependent $\rho(T)$, $C_p(T)$ on the frozen conductor cross-section to give the temperature trajectory $T(t)$ and the MITs budget 3.672 MA 2 s. The fused-detector statistics of equation (12) were evaluated on the modelled precursor transient, returning the channel trigger times and the fused signal-to-noise. The in-winding NV-diamond magnetometer that supplies the magnetization channel is specified and its neutron functional-life margin computed in the dedicated companion study [37]; here the sensor enters through the frozen $39\times$ life margin.

3.4 RAMI Monte-Carlo

The availability of equation (13) was evaluated (gate 04) with a time-domain reliability model of the demountable-cartridge swap: uptime intervals set by τ_{cp} , swap downtime distributions, and an overlap fraction β for the merge with scheduled maintenance. The dedicated-versus-merged availability figures follow directly; the full fleet-level time-domain Monte-Carlo across a four-unit breeder fleet is developed in the companion RAMI study [38].

4 Results

4.1 The coverage-versus-protection trade

Table 1 gives the trade curve at the design point with the breeding mantle on, and figure 1 plots it. Net TBR falls monotonically from 1.196 (5 cm shield) to 1.097 (40 cm) as centrepost life at the Fischer threshold rises from 0.0048 to 0.194 fpy. The mantle-off reference ($f_{\text{in}} = 0$) gives net TBR 1.053 at life 0.206 fpy. The trade is steep and binding: no split simultaneously delivers a high TBR and a multi-year centrepost, which is precisely why the consumable frame is the right one. The

design picks a breeding-favourable split and absorbs the life as a scheduled swap. Figure 2 shows the same data as the breeding-versus-life Pareto set; the operating choice is a point on this boundary, not a compromise off it.

Table 1. Centrepost coverage-versus-protection trade at the 9.66 MA design point, breeding mantle on. Life columns are at the 1×10^{22} , 3.9×10^{22} (Fischer), and $1 \times 10^{23} \text{ n m}^{-2}$ thresholds; mid- current (6.57 MA) lives are $2.25 \times$ the design values at every split (equation 5). Source: gate BR-L1-A13b, `cp_split_scan_results.json`.

t_{sh} (cm)	mantle (cm)	net TBR	life 1e22	life 3.9e22	life 1e23	dpa/fpy	heat (W)
5	42	1.196	0.0012	0.0048	0.012	0.412	937
10	37	1.193	0.0018	0.0071	0.018	0.264	563
15	32	1.188	0.0029	0.0115	0.029	0.164	337
20	27	1.181	0.0051	0.0198	0.051	0.094	185
25	22	1.170	0.0086	0.0336	0.086	0.056	112
30	17	1.154	0.0155	0.0606	0.155	0.031	61
35	12	1.130	0.0279	0.109	0.279	0.018	36
40	7	1.097	0.0496	0.194	0.496	0.010	20
ref ($f_{\text{in}}=0$)	17	1.053	0.053	0.206	0.527	0.010	18

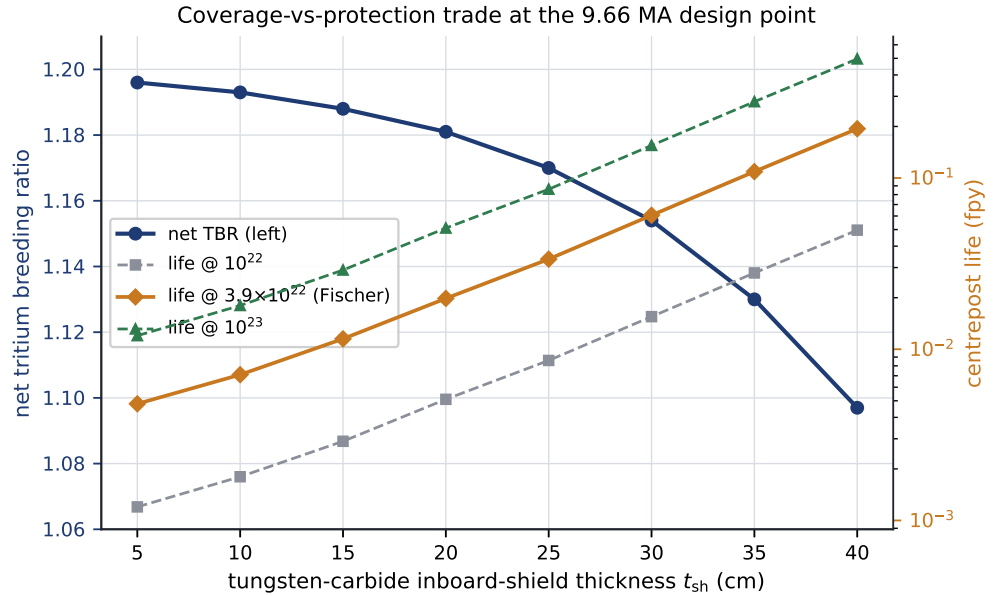


Figure 1. Coverage-versus-protection trade at three fluence thresholds (gate BR-L1-A13b): net TBR (navy, left) falls monotonically as the tungsten-carbide shield thickens, while centrepost life (right, logarithmic) rises. The amber diamonds are the Fischer onset threshold ($3.9 \times 10^{22} \text{ n m}^{-2}$), the binding retirement criterion.

4.2 Shield effectiveness: damage and cryogenic heat

The shield does more than protect the critical current: it attenuates the displacement-damage rate of equation (6) and the volumetric nuclear heating that the cryopant must remove. Figure 3 shows both falling quasi-exponentially with t_{sh} , with e-folding lengths of $\sim 11 \text{ cm}$ for the damage rate and $\sim 9 \text{ cm}$ for the heating. Across the scan the damage rate falls from 0.412 to 0.010 dpa/fpy and the centrepost nuclear heating from 937 to 20 W — a factor of ~ 47 in the cold- mass heat load, which is the coupling into the cryogenic budget quantified in the companion cold-mass study [43]. The shield choice is therefore a three-way coupling of breeding, magnet life, and refrigeration load, all resolved on the single variable t_{sh} .

4.3 The iso-elastic mission dial

Equation (5) makes the life a geometry-only quantity scaled by the inverse source strength. Figure 4 plots the Fischer-threshold life for the 9.66 MA design point and the 6.57 MA mid-current point, which differ everywhere by the exact factor 2.25. At the most-shielded end of the trade — a full 45 cm tungsten-carbide shield — the best-case cartridge life reaches 0.58 fpy, the scheduled-replacement design point. The mission dial is thus a genuine operational lever: derating the current (or, more precisely, the source strength) buys centrepost life on a one-for-one inverse basis, and the

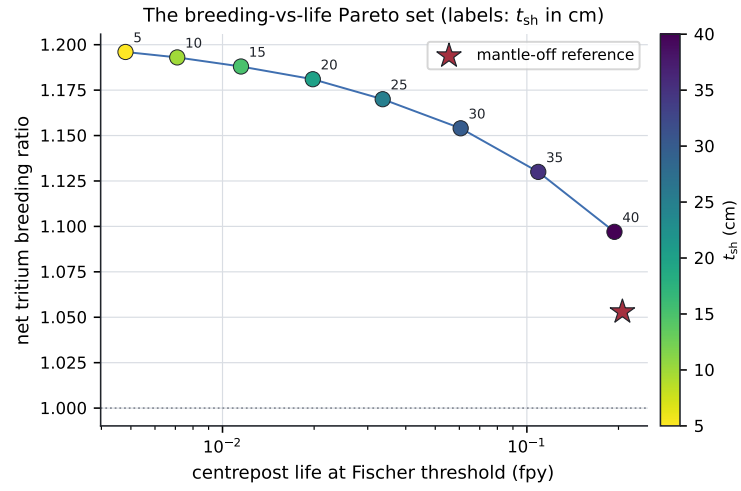


Figure 2. The breeding-versus-life Pareto set from Table 1; marker colour and label give the shield thickness t_{sh} (cm), the abscissa is centrepost life at the Fischer threshold (logarithmic), and the red star is the mantle-off reference. Every attainable operating point lies on this frontier.

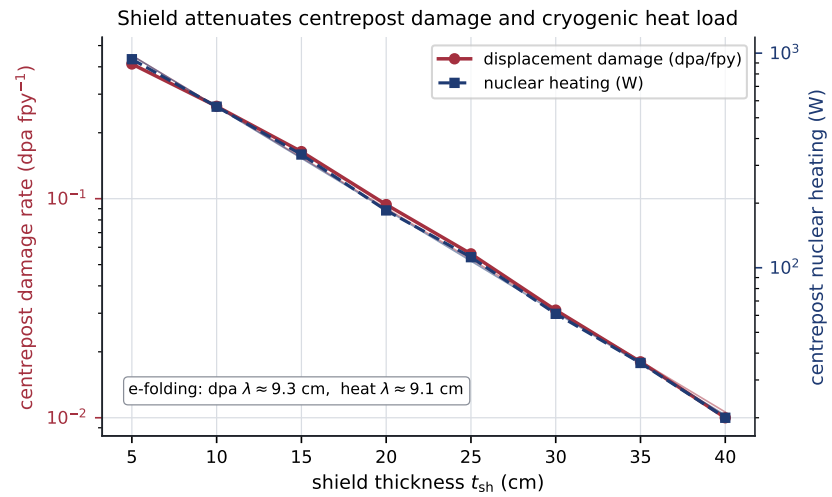


Figure 3. Shield attenuation of centrepost displacement damage (red, dpa/fpy) and nuclear heating (navy, W) versus t_{sh} ; both axes are logarithmic and the faint lines are single-exponential fits with the quoted e-folding lengths. The heat curve is the coupling into the cryogenic budget (gates BR-L1-A13b, BR-L1-A13).

admissible current window that results from folding this against the source-strength requirement is the subject of the companion operating-window synthesis [34].

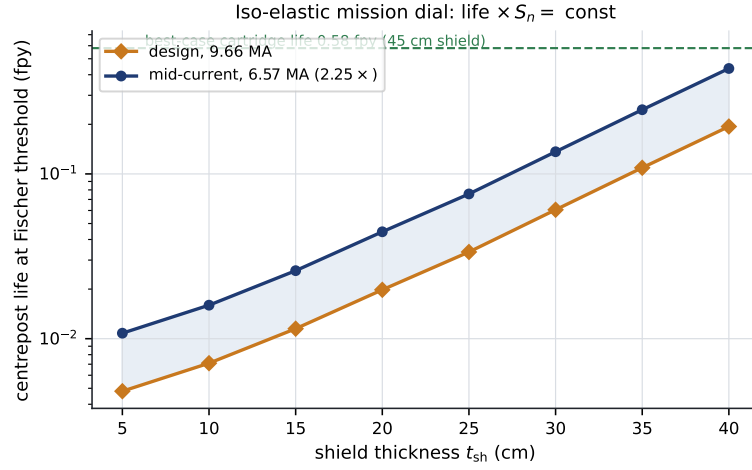


Figure 4. Iso-elastic mission dial (equation 5): centrepost life at the Fischer threshold for the design (9.66 MA) and mid-current (6.57 MA) points, separated everywhere by the exact factor 2.25. The dashed line marks the best-case 0.58 fpy cartridge life at the fully shielded (45 cm) end.

4.4 Coil structural integrity

Evaluated on the as-built centrepost, the closed-form bound of equation (9) gives a bore membrane hoop stress of 656.6 MPa, and the wedged-sector finite-element solution gives 625.3 MPa — a difference of -4.75% . Figure 5 plots the full radial profile: the finite-element and Lamé curves track each other across the whole bore-to-outer span, peaking at the bore and falling monotonically outward, and agreeing to within a few per cent everywhere. The agreement verifies the reduced-order structural model and confirms the membrane hoop stress as the binding structural quantity. The monotone profile motivates a per-turn engineered residual pre-compression that flattens the Lorentz strain gradient — the mechanical companion to the neutronics-driven swap cadence, and consistent with the structural mechanics of the surrounding vessel support [44].

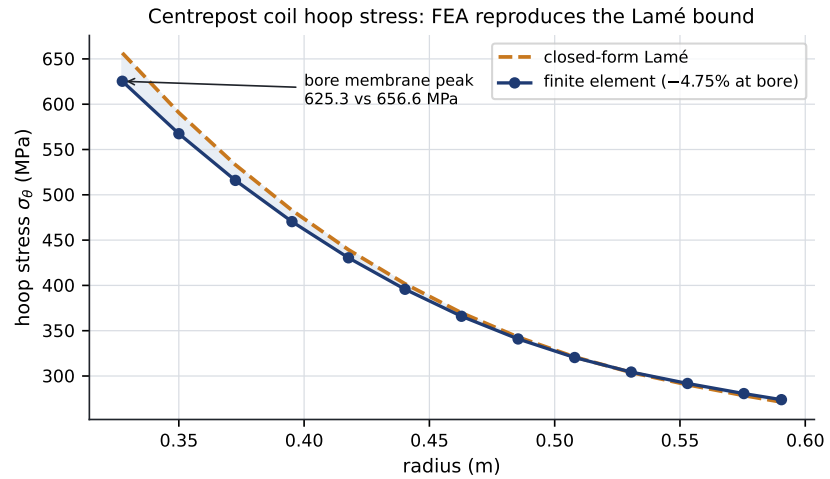


Figure 5. Bore-resolved centrepost coil hoop stress: wedged-sector finite element (navy) versus the closed-form Lamé thick-wall bound (amber, equation 9), agreeing to -4.75% at the bore membrane peak (625.3 versus 656.6 MPa; gate 02). The shaded band is the FEA-analytic difference.

4.5 Quench protection and fused detection

Integrating the adiabatic hot-spot balance of equations (10)–(11) gives the conductor temperature trajectory of figure 6. The fused strain-rate-plus-magnetization detector triggers at 0.089 s, when the conductor has reached only 47 K, whereas conventional voltage detection does not fire until 3.024 s, at 174 K. The protection lead is 2.935 s, comfortably inside the 3.672 s MITs budget, and

the fused window statistic reaches a signal-to-noise of 322.5. Figure 7 shows why fusion wins: the strain-rate and magnetization channels carry window-statistic signal-to-noise of 2.00 and 2.78, and their quadrature combination (equation 12) yields 3.43 — a strictly larger statistic than either channel, with the earliest trigger time. The magnetization channel is supplied by an in-winding NV-diamond magnetometer whose coherence survives the alpha environment; fast-neutron creation of NV centres sets a functional-life requirement that is met with a factor-39 margin over one cartridge life (0.22 fpy design, 0.496 fpy mid-current), the same margin carried in the dedicated NV-quench study [37].

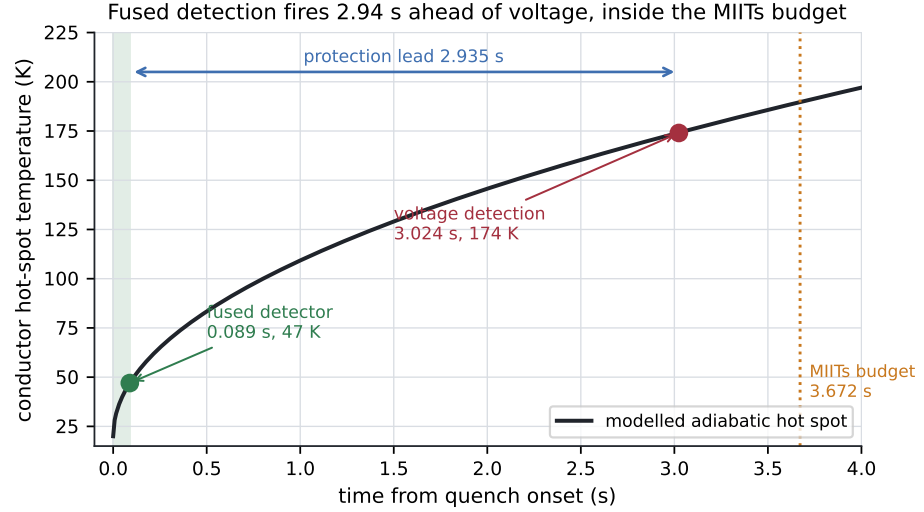


Figure 6. Modelled adiabatic hot-spot temperature trajectory (equations 10–11; gate 03). The fused detector fires at 0.089 s / 47 K (green), voltage detection at 3.024 s / 174 K (red); the protection lead of 2.935 s sits inside the 3.672 s MIITs budget (amber). The connecting curve is the model; the two detection points are the frozen results.

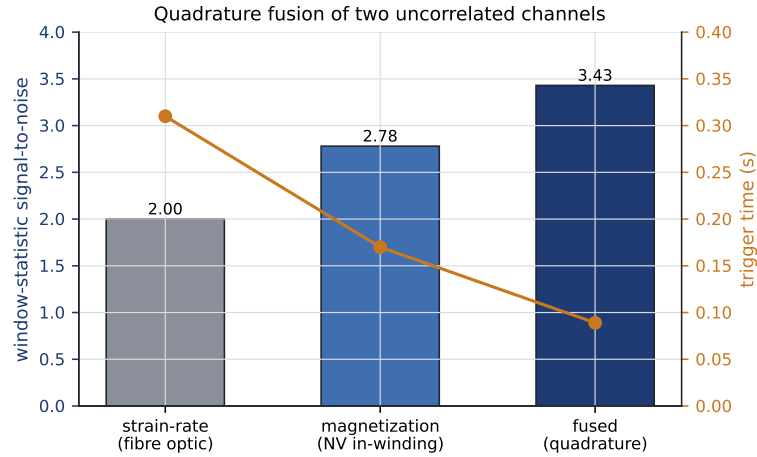


Figure 7. Quadrature fusion of two uncorrelated precursor channels (equation 12; gate 03): window- statistic signal-to-noise for the strain-rate, magnetization, and fused channels (bars, left axis) and their modelled trigger times (amber, right axis). Fusion produces both the largest statistic and the earliest trigger.

4.6 Maintenance cadence and availability

The demountable cartridge turns the finite centrepost life into a swap schedule. Evaluating the renewal model of equation (13) on the frozen swap logistics gives a plant availability of 0.5825 when the centrepost swap is a dedicated outage, rising to 0.5904 when swaps are merged with otherwise-scheduled maintenance (figure 8). At the longer mid-current cadence the same model gives 0.6382 (dedicated) and 0.6477 (merged). We report this honestly: all four figures sit below a conventional continuous-duty reference (capacity factor 0.69), and the merge — removing the swap’s dedicated contribution to the downtime denominator — is the single largest controllable lever on availability. Availability, not breeding, is the line item that the consumable-centrepost

architecture must manage; the fleet-level time-domain treatment across a four-unit breeder fleet is the subject of the companion RAMI study [38].

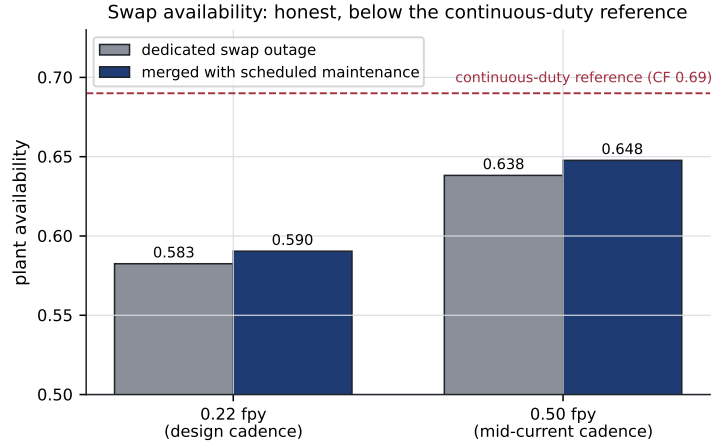


Figure 8. Plant availability from the renewal model (equation 13; gate 04): dedicated swap outage (grey) versus merge with scheduled maintenance (navy), at the design (0.22 fpy) and mid-current (0.50 fpy) cadences. All figures sit honestly below the continuous-duty reference (dashed, CF 0.69).

4.7 The architecture, assembled

Figure 9 is the radial-build schematic that ties the four analyses together: the conductor, the scan-variable shield, and the breeding mantle share one 72 cm inboard column, and figure 10 is the fused-detection architecture that protects it. Together they realise the consumable-centrepost concept — a demountable cartridge whose life is set by neutronics, whose structure is verified against a closed form, whose quench is caught by a fused non-voltage detector, and whose swap is costed in availability.

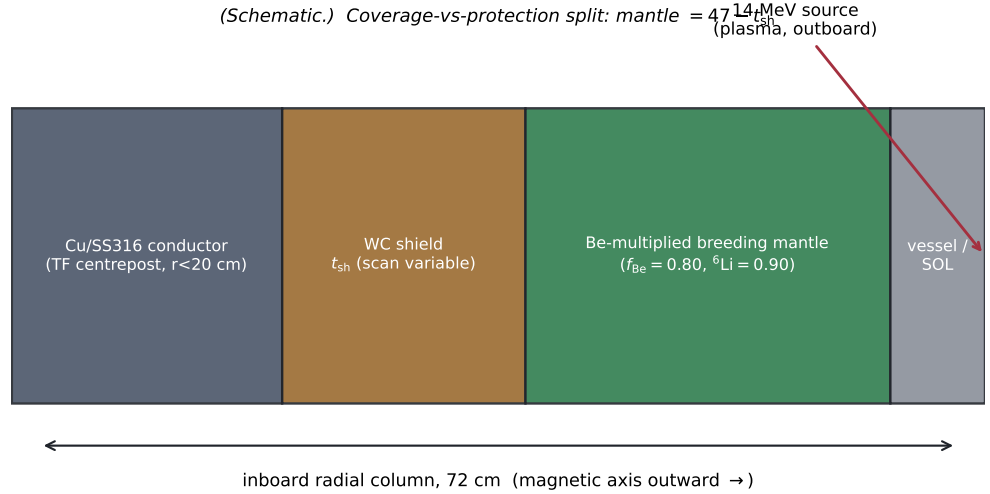


Figure 9. (Schematic.) The consumable-centrepost inboard radial build: Cu/SS316 conductor ($r < 20$ cm), tungsten-carbide shield of scan-variable thickness t_{sh} , and beryllium-multiplied breeding mantle ($47 - t_{sh}$), sharing one 72 cm column with the 14 MeV source outboard. No data values are shown; the figure encodes the coverage-versus-protection split only.

5 Discussion

The four analyses close a consistent picture, and each admits a quantitative comparison with the published literature.

Shielding and breeding. Our tungsten-carbide centre-column shield attenuates the conductor fast flux strongly enough that the binding constraint becomes centrepost life rather than breeding, exactly the regime Windsor and co-workers identified for tungsten-boride shields in a compact ST,

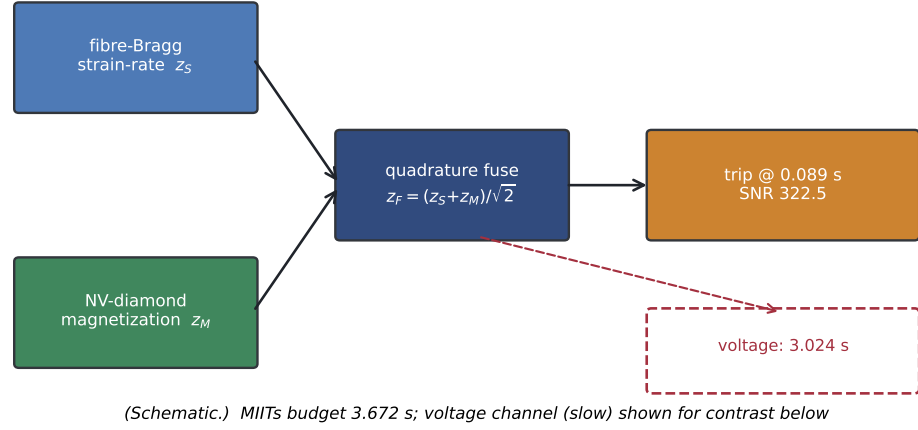


Figure 10. (Schematic.) The fused quench-detection architecture: a fibre-Bragg strain-rate channel and an in-winding NV-diamond magnetization channel are combined in quadrature (equation 12) to trip at 0.089 s, far ahead of the slow voltage channel (3.024 s). Structure only; the numeric labels are the frozen detection results.

where flux and heating are cut by about an order of magnitude relative to bare tungsten [19, 18]. Our net TBR stays above unity across the entire scan (1.097–1.196), which places the machine on the self-sufficient side of the breeding requirement documented for compact DEMO-class systems [23, 5]; the fleet-level tritium export case rests on the same net-TBR result [40, 41]. The steepness of the trade — a $\sim 40\times$ swing in life for a $\sim 8\%$ swing in TBR — is what forces the consumable frame: there is no split that is simultaneously breeding-rich and long-lived.

The REBCO fluence limit. The Fischer onset threshold of $3.9 \times 10^{22} \text{ n m}^{-2}$ [12] that sets the retirement criterion is consistent with the broader reactor-irradiation literature on coated conductors, which reports heavy in-field critical-current degradation and transition-temperature suppression at fluences of this order [13]. Adopting the *onset* of degradation rather than a fractional-loss criterion is deliberately conservative and keeps the design well up the plateau of equation (7); the first-principles justification — oxygen-vacancy formation energies and displacement cascades in YBCO — is developed in the companion radiation-tolerance study [35], and the conductor and stabiliser materials response under irradiation in the conductor-materials study [36].

Coil structure. The -4.75% agreement between the wedged-sector finite element and the closed-form Lamé bound (equation 9) is well within the tolerance expected for a homogenised-winding-pack membrane calculation, and the bore-peaked monotone profile matches the classical thick-wall solution [30]. The 625.3 MPa bore stress is a large but manageable membrane load for a stainless-steel-reinforced winding pack, consistent with the structural envelopes reported for high-field REBCO magnets [8, 11]. Crucially, the breeder centrepost governs at 16.84 T; it is a distinct device from the separate 26.49 T burner plug coil [39], and the two must never be conflated.

Quench detection in context. The fused detector should be read against the slow-normal-zone-propagation problem that motivates it. Marchevsky’s review makes the core point quantitatively: in HTS conductors the large temperature margin and slow propagation mean a hot spot can reach damaging temperature before a measurable resistive voltage appears, so voltage detection alone is inadequate [24]. The community response has been non-voltage observables: Salazar *et al.* demonstrated fibre-Bragg thermometry on VIPER cable at SULTAN with signal-to-noise of 4–32 for heater-induced events [25], and Teyber *et al.* showed current-distribution monitoring detecting quench and damage in fusion cables [26]. Our contribution is complementary: fusing an uncorrelated strain-rate and magnetization pair (equation 12) lifts the window statistic above either channel and buys a 2.935 s protection lead inside the MIITs budget — early, low-false-alarm warning in the low-signal corner the slow-propagation physics forces a real magnet into. The picotesla-class sensitivity of the in-winding NV magnetometer sits well inside the envelope established for ensemble NV magnetometry [27, 28, 29], so the sensing claim rests on a conservative device with large margin for multiplexing and in-service degradation.

Availability. The reported availabilities (0.5825–0.6477) are honestly below a continuous-duty

reference, and we do not dress them up. The value of the analysis is that it makes the availability penalty of the consumable centrepost *explicit and controllable*: the merge lever recovers roughly one percentage point, and the mid-current cadence lever recovers several more, both of which are quantified against the same renewal model [38]. This is the standard RAMI posture for maintenance-limited fusion systems [5], applied to a component whose finite life is treated as a scheduled interval rather than a failure.

6 Limitations

The results are design-and-simulation on the frozen configuration, not measurements from an operating machine. The one open engineering item is the in-winding NV sensor’s neutron tolerance: the analytical sensor sensitivity assumes an ensemble coherence and contrast that will degrade under the high background field, cryogenic operation, and fast-neutron creation of NV centres in the diamond. The mitigation is quantified — the functional-life requirement is met with a factor-39 margin over one cartridge life, and the fused detector retains its precursor advantage with the NV channel heavily degraded — but a dedicated NV neutron-tolerance study (irradiation plus first-principles charge-state modelling) is the follow-on that closes it [37]. Beyond that, the homogenised conductor composition and the wedged-sector mechanical model are reduced representations of a real winding pack; the fused-detector lead margin is computed for a modelled transient and awaits bench confirmation on real REBCO; and the availability figures depend on swap-logistics assumptions stated with the model rather than measured. The neutronics carry Monte-Carlo statistics (relative error 0.01–0.03) and the Fischer threshold is a retrieved onset value, not a device-specific measurement.

7 Conclusion

Treated as a consumable, the centrepost stops being a showstopper. On the frozen Hyperion design point the coverage-versus-protection trade is steep and binding — net TBR 1.196 to 1.097 against Fischer-threshold life 0.0048 to 0.194 fpy across the shield scan — so the design chooses breeding coverage and absorbs the life as a scheduled swap, with the damage rate and cryogenic heat load attenuated by $\sim 40\times$ and $\sim 47\times$ respectively and the life obeying an exact iso-elastic mission-dial law with a $2.25\times$ margin to mid-current. The coil matches its analytic hoop stress to -4.75% , a fused strain-rate-plus- magnetization detector leads voltage detection by 2.935 s inside the 3.672 s MIITs budget at signal-to-noise 322.5, and the swap costs availability (0.5825–0.6477) that the merge with scheduled maintenance partly recovers. The centrepost’s finite life becomes a maintenance line item with a quantified cadence, not a barrier to a compact ST breeder. Hyperion is a breeder and materials platform; these results describe how it survives its own neutron environment.

Acknowledgments

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Reproducibility. Every headline quantity in this paper is a frozen anchor in the Kronos Physics De-Risking Register [32] (doi:10.5281/zenodo.22133057): the coverage-versus-protection scan (gate BR-L1-A13b), the full-machine neutron balance (BR-L1-A5), the neutron-damage rate (BR-L1-A13), the coil finite- element regression (gate 02), the fused-detector timeline (gate 03), and the RAMI availability (gate 04). Each is regenerable from the archived solvers and the figure-generation script deposited with this paper; this paper is archived at doi:10.5281/zenodo.22645704; official version: <https://www.kronosfusionenergy.com/publications/centrepost-and-magnet-integrity.html>. No result depends on unpublished or proprietary data.

Scope — what this paper does not claim. The centrepost life is a scheduled-replacement interval, not a claim of a long-lived component; the availability figures are engineering projections, not operating data; and no net-electricity claim is made.

Data availability

The trade curve, the shield-attenuation series, the coil hoop-stress profile, the quench transient, and the RAMI figures used for figures 1–8 are archived with the deposit at

doi:10.5281/zenodo.22645704, which references the Kronos 2026 series including the breeder platform [31] and the De-Risking Register [32]. Headline quantities are regenerable by named scripts (`cp_split_scan.py`, `centrepost_fea.py`, `quench_analysis.py`) that read their inputs from a declared-parameter table. All quantitative values trace to the register by the gate identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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